Epi model and coalescent – Notes

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1 Basic model idea

Host: Fixed population size N, non-overlapping generations. Host infected or not infected, no effect on the fitness.

Pathogen:
First step is vertical transmission with probability c_v. In that, we establish a population of newborns with I_n infected (seedling infecteds) and $N - I_n$ uninfected individuals. The second step is an epidemiological process during the life span of the individuals of our focal generation. This process leads to more (or perhaps also less) infected individuals. At the end of the life time, X_n individuals are infected.

We do not assume that there is an external reservoir generating newly infecteds, but that we have an isolated population.

That is, we have two steps to obtain X_{n+1} from X_n:

$$I_{n+1} \sim \text{Binom}(N, c_v\, X_N/N)$$

gives us the number of infected newborns (founding infecteds of generation $n+1$).

I_{n+1} serves as the initial state of the epidemiological process J_t within/during generation $n+1$. We have N individuals altogether, $J_0 = I_{n+1}$ are initially infected, and $N - J_0$ are initially susceptible. The epidemiological process runs. We aim at the size distribution at time G as we have

$$X_{n+1} = J_G.$$

1.1 Heuristic approach for the SI model for N large

The choice of the epidemic model depends on the infection we have in mind. One simple choice might be an SI model. That is, if J_t is the number of infecteds at time t, we have $N - J_t$ susceptibles and

$$J_t \to J_t + 1 \quad \text{at rate } \beta(N - J_t)J_t/N.$$

Strictly spoken, we require the distribution of $J_G = I_G$. Anyhow, due to correlations this distribution is not that easy to evaluate.

For a simple approach, we assume that N as well as J_0 are large such that a deterministic approximation of the epidemic model is valid. If we define $i(t) = J_t/N$, then $i(t)$ approximately follows the ODE

$$i' = \beta i(1 - i), \qquad i(0) = i_0$$

with the solution

$$i(t) = \frac{i_0}{e^{-\beta t}(1 - i_0) + i_0}.$$

We might interpret $i(t)$ as the probability for an individual to be infectious. As the population size N is large, the correlations tend to zero, and approximately we can treat the individuals as independent.

With this assumptions, we find approximately

$$X_{n+1} = I_{n+1} + K_{n+1}$$

where I_{n+1} are the founding infecteds in generation $m+1$, and K_{n+1} the (during generation $n+1$) newly infecteds due to the epidemiological process,

$$\begin{aligned} I_{n+1} &\sim \text{Binom}(N, c_v\, X_N/N) \\ K_{n+1} &\sim \text{Binom}\left(N - I_{n+1},\ \frac{I_{n+1}}{e^{-\beta G}(N - I_{n+1}) + I_{n+1}}\right). \end{aligned}$$

1.1.1 Deterministic limit

If we take N to infinity, we loose stochastic effects, and with $x_n = X_n/N$ we have $I_{n+1}/N \approx c_v x_n$ and

$$x_{n+1} = c_v x_n + (1 - c_v x_n)\frac{c_v x_n}{e^{-\beta G}(1 - c_v x_n) + c_v x_n}.$$

The stationary solutions satisfy $x_{n+1} = x_n = x$,

$$x = c_v x + (1 - c_v x)\frac{c_v x}{e^{-\beta G}(1 - c_v x) + c_v x}$$

that is, either $x = 0$ (uninfected equilibrium), or

$$\begin{aligned} 1 &= c_v + (1 - c_v x)\frac{c_v}{e^{-\beta G}(1 - c_v x) + c_v x} \\ (1 - c_v x)\, c_v &= (1 - c_v)\left(e^{-\beta G}(1 - c_v x) + c_v x\right) = (1 - c_v)\, e^{-\beta G} + x\,(1 - c_v)\, c_v \left(1 - e^{-\beta G}\right) \\ c_v - (1 - c_v)\, e^{-\beta G} &= x\left\{c_v^2 + (1 - c_v)\, c_v \left(1 - e^{-\beta G}\right)\right\} \\ x &= x^* = \frac{c_v - (1 - c_v)\, e^{-\beta G}}{c_v^2 + (1 - c_v)\, c_v\ (1 - e^{-\beta G})} \end{aligned}$$

On the other hand, if x^* and c_v is known, we can infer $e^{-\beta G}$,

$$\begin{aligned} 1 &= c_v + (1 - c_v x^*)\frac{c_v}{e^{-\beta G}(1 - c_v x^*) + c_v x^*} \\ (1 - c_v x^*)\, c_v &= (1 - c_v)\left(e^{-\beta G}(1 - c_v x^*) + c_v x^*\right) \\ e^{-\beta G} &= \frac{(1 - c_v x^*)\, c_v - (1 - c_v)\, c_v x^*}{(1 - c_v)\,(1 - c_v x^*)} = \frac{c_v - c_v^2 x^* - (c_v x^* - c_v^2 x^*)}{(1 - c_v)\,(1 - c_v x^*)} = \frac{c_v}{(1 - c_v)}\ \frac{(1 - x^*)}{(1 - c_v x^*)}. \end{aligned}$$

Obviously, $c_v \in (0, c_v^*)$ where $c_v^* = c_v^*(x^*)$ is an upper bound for c_v that guarantees that the r.h.s. is smaller 1.

1.1.2 Coalescence probability for pathogen lineages

In order to construct coalescent trees for the pathogen lineages, it is not sufficient to know how many individuals are infected, but moreover how the infected individuals cluster within clades, where a "clade" is the set of infected individuals directly or indirectly infected by one initial founding infected.

We select randomly two pathogen lineages, and ask for the coalescence probability within one generation. There are two possibilities for an coalescence event: Either the lineages stem from the same clade (probability π), or they stem from different founding infecteds. In the latter case, the probability for coalescence is $1/N$, such that

$$p_c = P(\text{coalescence of two pathogen lineages within one host generation}) = \pi + (1 - \pi)\,\frac{1}{N}.$$

It is thus central to understand π.

Thereto we set up an urn model that generates the probability distribution of the clade sizes, given the total number of infecteds: We start with I balls, each of them has a different colour. Then we randomly draw a ball from the urn, and put that ball together with a further ball of the same colour back into the urn. This is a classical Pólya urn model. We derive π in the appendix (see eqn (1)), and obtain

$$\pi = \frac{j}{j+I-1}\,\frac{2}{I+1}.$$

For large numbers of additionally infecteds ($j \to \infty$), we have *(attention: Here we implicitly assume $I \ll j$, which might depend on the parameters and is perhaps not always justified!! We need here $c_v \ll 1$, which might not be unreasonable)*

$$\pi = \frac{2}{I+1}.$$

Note that this probability is larger than the first naive guess $1/I$, where we simply assume that each founding infected has the same probability to be the origin of the clade for a sampled infected individual. The deeper reason is that clades which initially grow a little faster by stochastic effects explode in size due to the exponential growth of clades. That is, we will have some very large, dominating clades. This effect increases the probability to draw two balls from the same clade.
Anyhow, the coalescence probability p_c for two randomly sampled infected lineages becomes

$$p_c = \pi + \frac{1-\pi}{N} = \frac{2}{I+1} + \frac{1-\frac{2}{I+1}}{N} = \frac{2}{I+1} + \frac{I+1-2}{(I+1)\,N} = \frac{2\,N+I-1}{(I+1)\,N}.$$

In the equilibrium, $I = c_v\,x^*\,N$, such that for $N \gg 1$

$$p_c = \frac{2\,N + c_v\,x^*\,N - 1}{(c_v\,x^*\,N+1)\,N} \approx \frac{2\,N + c_v\,x^*\,N}{(c_v\,x^*\,N+1)\,N} = \frac{2+c_v\,x^*}{1+c_v\,x^*\,N} \approx \frac{1}{N}\,\frac{2+c_v\,x^*}{c_v\,x^*} = \frac{1}{2N_e^i}$$

with the infected-effective population size

$$N_e^i = N\,\frac{c_v\,x^*}{4+2\,c_v\,x^*}.$$

Interpretation (*handle with care – not sure if the result is ok, nor if the interpretation is ok*): $N\,c_v\,x^*$ is the effective transmitting population (from one generation to the next one). This is the transmission process between the generations. This effective population size is further decreased by the denominator $4+2\,c_v\,x^*$, which reflects the increased coalescence probability due to the fact that some clades produced by the seedling infecteds will dominate (infection process within the generation).
Let A_n denote the number of pathogen lineages n generations back in time. Then, if N is large (that is, $p_c = \mathcal{O}(N^{-1}) \ll 1$)

$$P(A_{n+1} = A_n - 1 | A_n) \approx p_c \binom{A_{n-1}}{2} = \frac{1}{2\,N_e^i}\binom{A_{n-1}}{2}.$$

1.1.3 Estimation procedure

Take a sample of hosts, and sequence the host genome. Determine which hosts are infected, and sequence the pathogens.
Estimate the Watterson estimator for the population size N_e using mutations in the host genome.
Estimate the Watterson estimator for the infected-population size N_e^i using mutations in the pathogen genome.
Then, *check!!*

$$\frac{N_e^i}{N_e} = \frac{c_v\,x^*}{2+c_v\,x^*}.$$

From here, we can estimate $c_v x^*$. Additionally we infer x^* from the fraction of infected hosts in the sample. In that we also estimate c_v alone. Lastly, from x^* and c_v, we obtain $e^{-G\beta}$.

The variance of the estimator might be rather small, as the coalescence tree for the pathogen is nested in that for the host.

A Simple Polya urn

Well known results, but somehow I did not find a good reference (the promising books are not available as e-books, and I'm too lazy to go down to the library).

We consider the most simple Polya urn (also called the epidemic urn): We start with k balls, and in each move we draw a ball and return this ball together with a further ball of the same colour. After n steps, we have some configuration of the form

$$(1+x_1,..,1+x_k) \qquad x_i\mathbb{N}_0, \qquad \sum_{\ell=1}^k x_i = n.$$

Proposition A.1. *Let $k>1$. All possible configurations $(1+x_1,..,1+x_k)$, $x_i\in\mathbb{N}_0$, $\sum_{\ell=1}^k x_i = n$ have the same probability*

$$p_n = \frac{1}{\binom{n+k-1}{n}}.$$

Proof: We use induction. For $n=0$, we only have one configuration $(1,..,1)$, and indeed

$$p_0 = \frac{1}{\binom{0+k-1}{0}} = 1.$$

Now we go from n to $n+1$. Let $x_1+..+x_k=n$. Then,

$$\begin{aligned} P(1+x_1,..,1+x_i+1,..,1-x_k) &= \frac{1+x_i}{k+n}P(1+x_1,..,1+x_i,..,1-x_k) \\ &\quad + \sum_{\ell\neq i}\frac{1+x_\ell-1}{k+n}P(1+x_1,..,1+x_i,..1+x_\ell-1,.,1-x_k) \end{aligned}$$

Note that this formula is correct also if $x_m=0$ for some m, as in that case $\frac{1+x_m-1}{k+n}=0$. We proceed

$$\begin{aligned} P(1+x_1,..,1+x_i+1,..,1-x_k) &= \frac{p_n}{k+n}\left(1+x_i+\sum_{\ell\neq i}x_\ell\right) = p_n\,\frac{(1+n)}{k+n} \\ &= \frac{1}{\binom{n+k-1}{n}}\,\frac{(1+n)}{k+n} = \frac{n!(k-1)!}{(n+k-1)!}\,\frac{(1+n)}{k+n} = \frac{(n+1)!(k-1)!}{(n+1+k-1)!} = p_{n+1}. \end{aligned}$$

□

After n Polya-Steps, we draw two balls without replacement, and ask for the probability that they have the identical colour.

Proposition A.2. *The probability q to draw from the Polya-Urn defined above after n Polya-steps two balls (without replacement) of the same colour reads*

$$q = \frac{n}{n+k-1}\,\frac{2}{k+1} \tag{1}$$

Proof Given the configuration $C=(1+x_1,..,1+x_k)$, $x_1+..+x_k=n$, we have the probability for our event E

$$P(E|C) = \sum_{\ell=1}^k\frac{(1+x_\ell)\,x_\ell}{(n+k)(n+k-1)} = \sum_{\ell=1}^k\frac{x_\ell^2+x_\ell}{(n+k)(n+k-1)} = \frac{n}{(n+k)(n+k-1)} + \frac{1}{(n+k)(n+k-1)}\sum_{\ell=1}^k x_\ell^2.$$

As each configuration C (for n Polya-steps) has the same probability, we obtain

$$P(E) = \frac{n}{(n+k)(n+k-1)} + \frac{1}{(n+k)(n+k-1)\binom{n+k-1}{n}}\sum_{x_1+..+x_k=n}\sum_{\ell=1}^k x_\ell^2.$$

Let $S = \sum_{x_1+..+x_k=n} \sum_{\ell=1}^k x_\ell^2$. We use $x_\ell^2 = 2\binom{x_\ell}{2} + x_\ell$,

$$S = n\binom{n+k-1}{n} + 2 \sum_{x_1+..+x_k=n} \sum_{\ell=1}^k \binom{x_\ell}{2} = n\binom{n+k-1}{n} + 2\sum_{\ell=1}^k \sum_{x_1+..+x_k=n} \binom{x_\ell}{2}.$$

For the last term, we find

$$\sum_{x_1+..+x_k=n} \binom{x_\ell}{2} = \sum_{x_1+..+x_k=n} \binom{x_1}{2} = \sum_{x_1=0}^n \binom{x_1}{2} \sum_{x_2+..+x_k=n-x_1} 1 = \sum_{\ell=0}^n \binom{\ell}{2}\binom{n+k-\ell-2}{n-\ell} = \binom{n+k-1}{n-2}$$

(last step via an llm, but checked numerically – needs a proof) such that

$$S = n\binom{n+k-1}{n} + 2\,k\binom{n+k-1}{n-2}.$$

Therewith,

$$\begin{aligned}
P(E) &= \frac{n}{(n+k)(n+k-1)} + \frac{1}{(n+k)(n+k-1)\binom{n+k-1}{n}} \sum_{x_1+..+x_k=n} \sum_{\ell=1}^k x_\ell^2 \\
&= \frac{n}{(n+k)(n+k-1)} + \frac{1}{(n+k)(n+k-1)\binom{n+k-1}{n}} \left(n\binom{n+k-1}{n} + 2\,k\binom{n+k-1}{n-2} \right) \\
&= \frac{2n}{(n+k)(n+k-1)} + \frac{2k}{(n+k)(n+k-1)} \frac{\binom{n+k-1}{n-2}}{\binom{n+k-1}{n}}
\end{aligned}$$

With

$$\frac{\binom{n+k-1}{n-2}}{\binom{n+k-1}{n}} = \frac{(n+k-1)!}{(n-2)!(k+1)!} \frac{n!(k-1)!}{(n+k-1)!} = \frac{n(n-1)}{(k+1)k}$$

we find

$$P(E) = \frac{2n}{(n+k)(n+k-1)} + \frac{2n}{(n+k)(n+k-1)} \frac{n-1}{k+1} = \frac{2n}{(n+k)(n+k-1)} \frac{n+k}{k+1} = \frac{n}{n+k-1} \frac{2}{k+1}.$$

$\square$